ITM 112 Team Project
Connected Systems
[100 Points]

1. Requirements

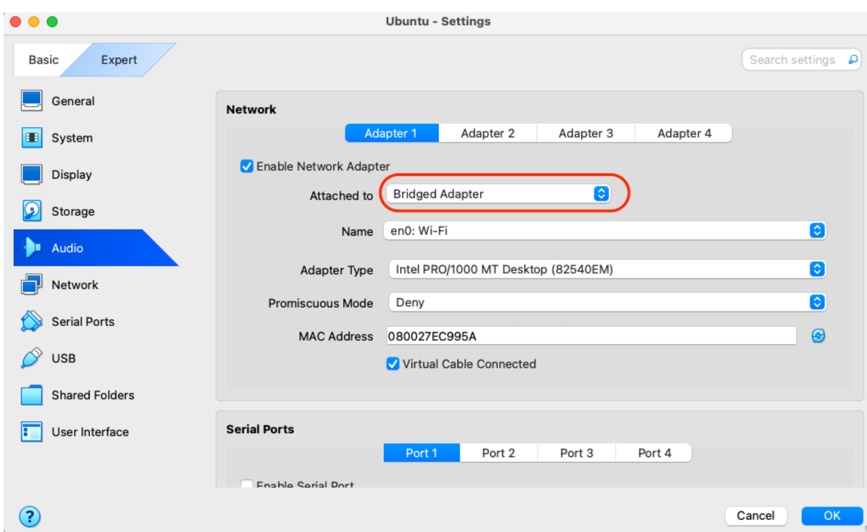
The virtual machine you created on your notebook computer can connect to other students' virtual machines to build a networked system. In this project, your team will build a system of connected servers. Each person in your team has the following assignment:

Team Member	Name	Server Deployment	Level of Difficulty
1		Apache web server on a VM	Easiest
2		Nginx web server on a Docker container	Moderate
3		Samba file share	More involved
4		A bash script for testing	Moderate

Use AI to learn commands and solve any issues you encounter.

2. Connectivity [Earn 20 points upon completion]

Your teammates' VMs must be in the same network (e.g., do it at school). VMs' network adapter must be set to "Bridged Adapter" (see the figure below).



The first step is to ping each other's VMs. Insert below a screen capture of the successful pings from one person's VM to all other VMs.

3. Apache Web Server on a VM [Earn 20 points upon completion]

Ask AI to generate an index.html file for you. Copy the index.html file to the server's html directory. Insert below a screen capture of the web page which is accessed from a web browser running on a remote VM.

4. Nginx Web Server on a Container [Earn 20 points upon completion]

Run an Nginx web server in a Docker container. Ask AI to generate an index.html file for you which is different from the file created for Apache web server. Copy the index.html file to the server's html directory. Insert below a screen capture of the web page which is accessed from a web browser running on a remote VM.

5. Samba File Share [Earn 20 points upon completion]

Server Side: To allow your teammates to access your Samba file share, you will need to add their user accounts in your Ubuntu (useradd -m). You will then be able to add them as Samba user (smbpasswd). Share /mnt/smb_share directory through Samba. Add a file named "samba.log" which contains the message, "*Yourname* is sharing this file with my teammates." Insert below a screen capture of the output of the following commands:

```
$ sudo sudo pdbedit -L; ls -l /mnt/smb_share
```

Client Side: At all other VMs, mount the file share at ~/smb. Add a file named "*yourname*.log" which contains the message, "*Yourname* is sharing this file with my teammates." Insert below a screen capture of the output of the following command from one of the client VMs:

```
$ mount | grep cifs; ls -l ~ | grep smb; ls -l ~/smb; cat
~/smb/*.log
```

6. Bash Script for Testing [Earn 20 points upon completion]

Write a bash script "itm112.sh" that performs the following operations. For each step, print a message to the console so that a progression of the testing is visible:

- A successful ping to all VMs run by your teammates (IP addresses can be hard coded)
- A successful downloading of index.html file from Apache web server using curl
- A successful downloading of index.html file from Nginx web server using curl

- A successful mounting of type cifs volume
- A successful listing of ~/smb/smb.log file (created by the samba share volume owner)
- A successful creation of ~/smb/bash.log file (empty file)

A sample output of the bash script is given below (All servers are running on localhost in this example. Yours should not be.):

```
[redacted]:~/scripts$ . itm112.sh
###
### Ping hosts
###
ping -c 1 -w 2 127.0.0.1 > /dev/null 2>&1
SUCCESS: Apache host 127.0.0.1 is rechable

ping -c 1 -w 2 127.0.0.1 > /dev/null 2>&1
SUCCESS: nginx host 127.0.0.1 is rechable

ping -c 1 -w 2 127.0.0.1 > /dev/null 2>&1
SUCCESS: Samba host 127.0.0.1 is rechable

###
### Test Web Servers
###
curl http://127.0.0.1 > /dev/null 2>&1
SUCCESS: Apache web server 127.0.0.1 is in service

curl http://127.0.0.1:8080 > /dev/null 2>&1
SUCCESS: nginx web server 127.0.0.1 is in service

###
### Test samba file share
###
mount | grep cifs > /dev/null 2>&1
SUCCESS: cifs volume is mounted

ls -l ~/smb/smb.log
-rwxrwxrwx 1 syslog okuda 61 Mar 25 22:31 /home/okuda/smb/smb.log
SUCCESS: ~/smb/smb.log exists

touch ~/smb/bash.log
SUCCESS: ~/smb/bash.log was created
```

Paste the screen captures of the execution of the bash script below.

Paste the content of the bash script in text below.